

JavaScript me loops ke **main 5 types** hote hain

JavaScript Loop Types:

1. **for loop**

- Fixed number of times chalane ke liye.
- Example: `for(let i=0; i<5; i++) { }`

2. **while loop**

- Jab tak condition true ho tab tak chale.
- Example:

```
while(condition) { }
```

3. **do...while loop**

- Pehle ek baar loop chalega, fir condition check hoga.
- Example:

```
do { } while(condition);
```

4. **for...of loop**

- Arrays ya iterable objects ke values ko access karne ke liye.
- Example:

```
for(let val of array) { }
```

5. **for...in loop**

- Objects ke properties (keys) ko access karne ke liye.
- Example:

```
for(let key in object) { }
```

code + expected output

1. for loop

```
for(let i = 1; i <= 5; i++) {  
    console.log("For loop:", i);  
}
```

Output:

```
For loop: 1  
For loop: 2  
For loop: 3  
For loop: 4  
For loop: 5
```

2. while loop

```
let i = 1;  
  
while(i <= 5) {  
    console.log("While loop:", i);  
    i++;  
}
```

Output:

```
While loop: 1  
While loop: 2  
While loop: 3  
While loop: 4  
While loop: 5
```

3. do...while loop

```
let j = 1;  
  
do {  
    console.log("Do-While loop:", j);  
    j++;  
} while(j <= 5);
```

Output:

```
Do-While loop: 1  
Do-While loop: 2  
Do-While loop: 3  
Do-While loop: 4  
Do-While loop: 5
```

4. for...of loop (Array ke elements par)

```
const fruits = ["apple", "banana", "mango"];  
  
for (const fruit of fruits) {  
    console.log("For...of loop:", fruit);  
}
```

Output:

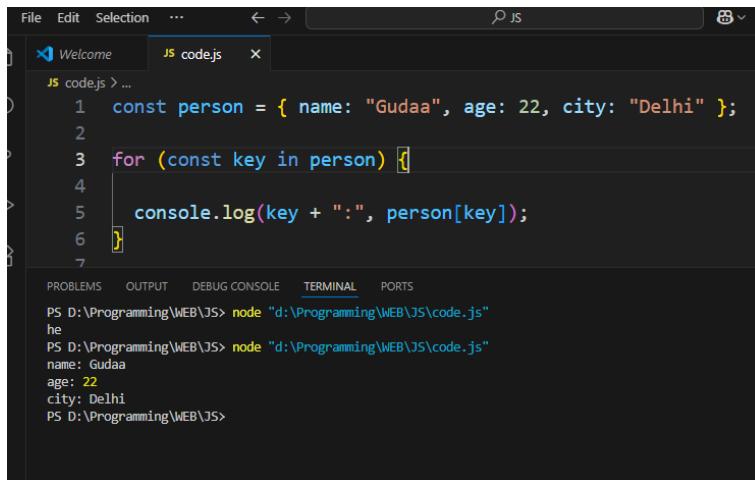
```
For...of loop: apple  
For...of loop: banana  
For...of loop: mango
```

5. for...in loop (Object ke properties par)

```
const person = { name: "Gudaa", age: 22, city: "Delhi" };  
  
for (const key in person) {  
    console.log(key + ":", person[key]);  
}
```

Output:

```
name: Gudaa  
age: 22  
city: Delhi
```



```
File Edit Selection ... JS
JS code.js x
JS code.js > ...
1 const person = { name: "Gudaa", age: 22, city: "Delhi" };
2
3 for (const key in person) {
4   console.log(key + ":", person[key]);
5 }
6
7
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Programming\WEB\JS> node "d:\Programming\WEB\JS\code.js"
he
PS D:\Programming\WEB\JS> node "d:\Programming\WEB\JS\code.js"
name: Gudaa
age: 22
city: Delhi
PS D:\Programming\WEB\JS>
```

: break and continue

```
for(let i = 1; i <= 5; i++) {
  if(i === 3) continue; // Skip number 3
  if(i === 5) break;    // Stop loop when i is 5
  console.log("Loop control:", i);
}
```

Output:

```
Loop control: 1
Loop control: 2
Loop control: 4
```
